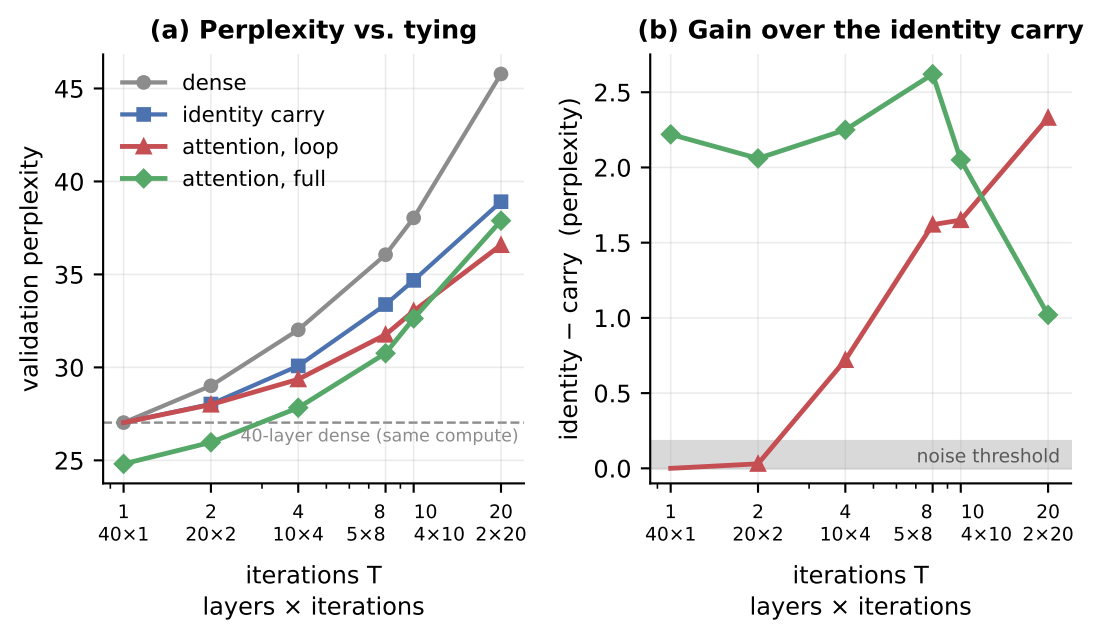
(a) Perplexity vs. tying
validation perplexity
45
40
35
30
25
dense
identity carry
attention, loop
attention, full
40-layer dense (same compute)
1
40×1
2
20×2
4
10×4
8
5×8
10
4×10
20
2×20
iterations T
layers × iterations
(b) Gain over the identity carry
identity − carry (perplexity)
2.5
2.0
1.5
1.0
0.5
0.0
noise threshold
1
40×1
2
20×2
4
10×4
8
5×8
10
4×10
20
2×20
iterations T
layers × iterations